Prediction Models Report

Cardiac Outcome Prediction using Dual Exponential Coefficients

Generated by Cardiac RODEO Pipeline

December 31, 2025

Abstract

This report presents the results of machine learning models trained to predict cardiac outcomes (Arrhythmia, Heart Damage, and Concern level) from drug response coefficients. Models were trained using Leave-One-Out Cross-Validation (LOOCV) for robust performance estimation.

Contents

1 Introduction

Three prediction models were trained on Dual Exponential coefficients:

- **Arrhythmia**: XGBoost classifier (binary: true/false)
- **Heart Damage**: RBF SVM classifier (binary: positive/negative)
- **Concern**: Random Forest classifier (multiclass: no/less/most)

All models were evaluated using Leave-One-Out Cross-Validation (LOOCV) to provide unbiased performance estimates on small sample sizes.

2 Model Performance

Table 1: Model Performance Metrics (LOOCV)

Model	Accuracy	F1-Score	AUC-ROC	Sensitivity	Specificity
Arrhythmia	0.6400	0.7097	0.7013	0.7857	0.4545
Heart Damage	0.8000	0.8889	0.2500	1.0000	0.0000
Concern	0.4400	0.2037	0.4241	–	–

Table 2: Concern Model Per-Class Metrics

Class	Accuracy	F1-Score	AUC-ROC
No	0.8400	0.0000	0.3452
Less	0.6000	0.0000	0.6404
Most	0.4400	0.6111	0.2867

3 ROC Curves

Figure ?? shows the ROC curves for all three models.

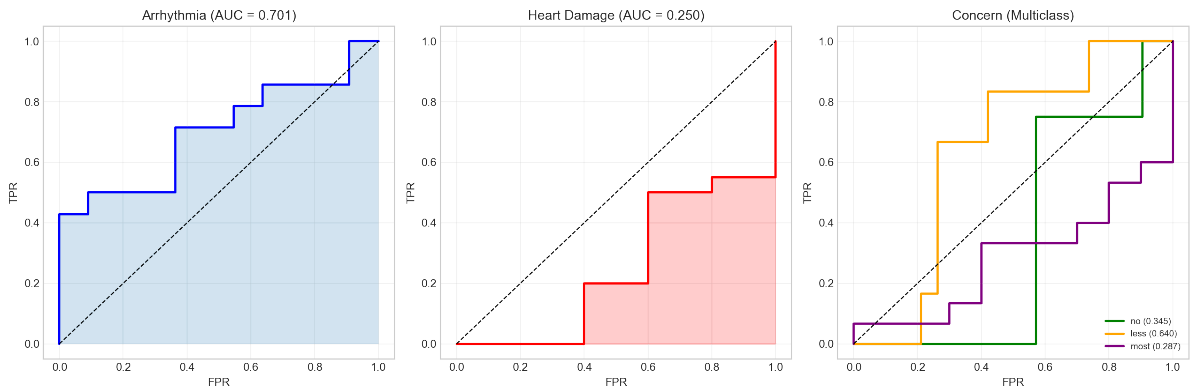

Figure 1: ROC curves for all prediction models.

3.1 Arrhythmia

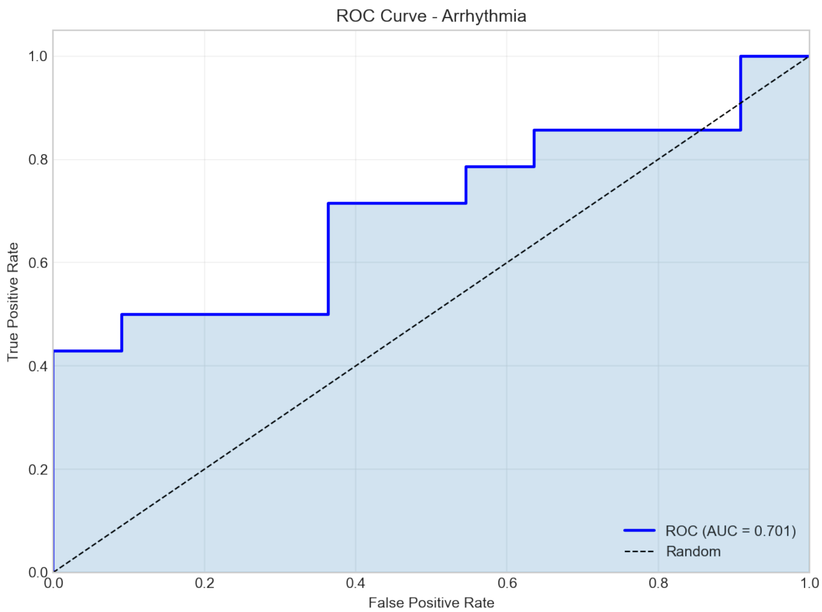

Figure 2: ROC curve for Arrhythmia prediction.

3.2 Heart Damage

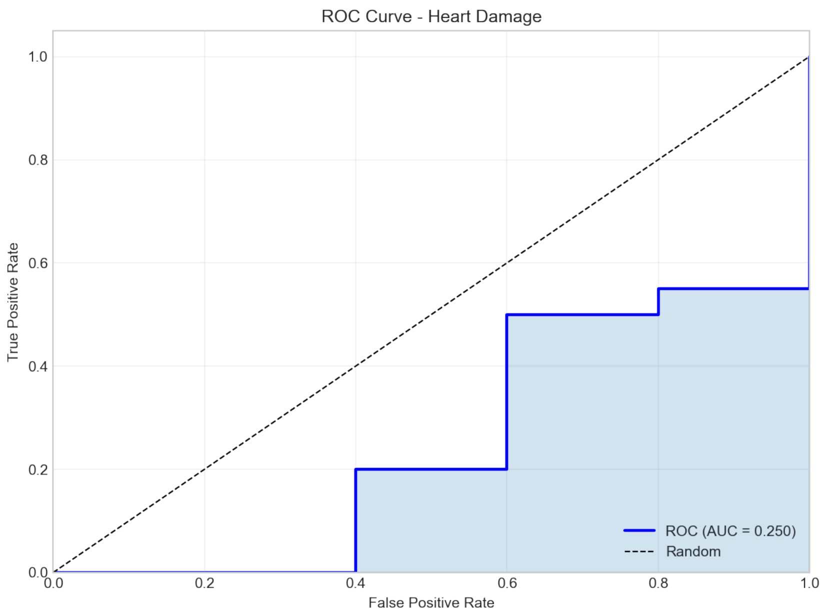

Figure 3: ROC curve for Heart Damage prediction.

3.3 Concern

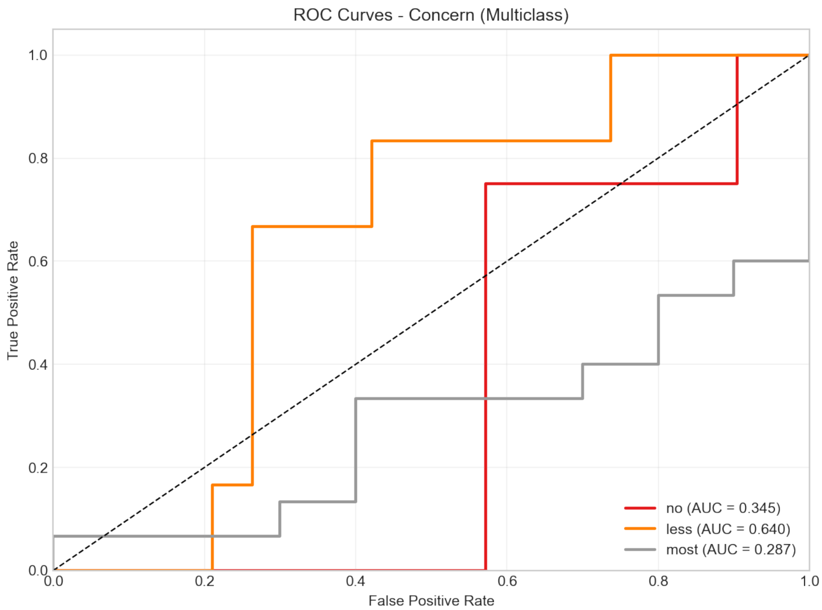

Figure 4: ROC curve for Concern prediction.

4 Confusion Matrices

4.1 Arrhythmia

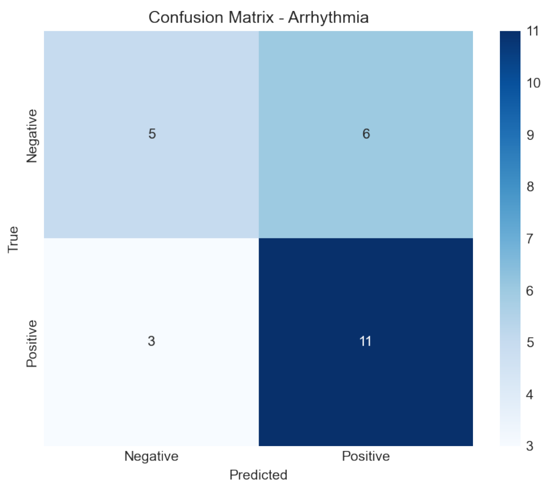

Figure 5: Confusion matrix for Arrhythmia prediction.

4.2 Heart Damage

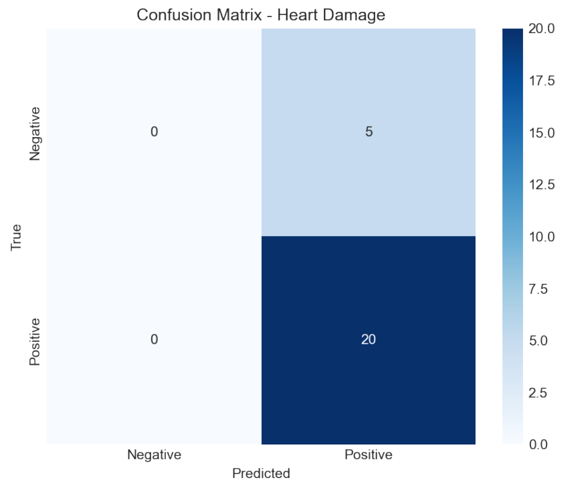

Figure 6: Confusion matrix for Heart Damage prediction.

4.3 Concern

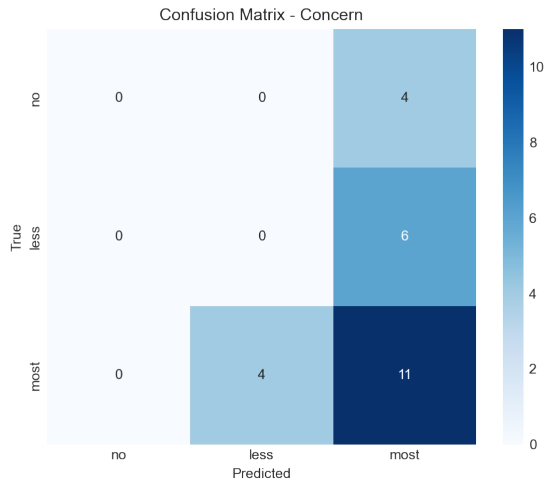

Figure 7: Confusion matrix for Concern prediction.

Table 3: Top 10 Most Important Features by Model

Rank	**Feature**	**Importance**
	Arrhythmia	
1	mt_Contractility	0.1955
2	R0_Contractility	0.1465
3	mb_Contractility	0.1218
4	tau_b_O2	0.0939
5	tau_b_Contractility	0.0864
6	A_tox_O2	0.0790
7	nb_Contractility	0.0776
8	mb_O2	0.0733
9	nt_Contractility	0.0527
10	tau_t_Contractility	0.0503
	Heart Damage	
1	kb_O2	0.0640
2	nb_O2	0.0600
3	nb_Contractility	0.0560
4	mb_O2	0.0520
5	nt_Contractility	0.0480
6	mt_O2	0.0440
7	tau_b_Contractility	0.0400
8	R0_Contractility	0.0400
9	A_tox_O2	0.0400
10	A_tox_Contractility	0.0360
	Concern	
1	A_benefit_Contractility	0.1089
2	nt_Contractility	0.1065
3	A_tox_Contractility	0.0733
4	tau_t_Contractility	0.0696
5	kt_O2	0.0582
6	mb_Contractility	0.0580
7	A_tox_O2	0.0567
8	tau_t_O2	0.0556
9	kb_O2	0.0529
10	tau_b_Contractility	0.0526

5 Feature Importance

5.1 Arrhythmia

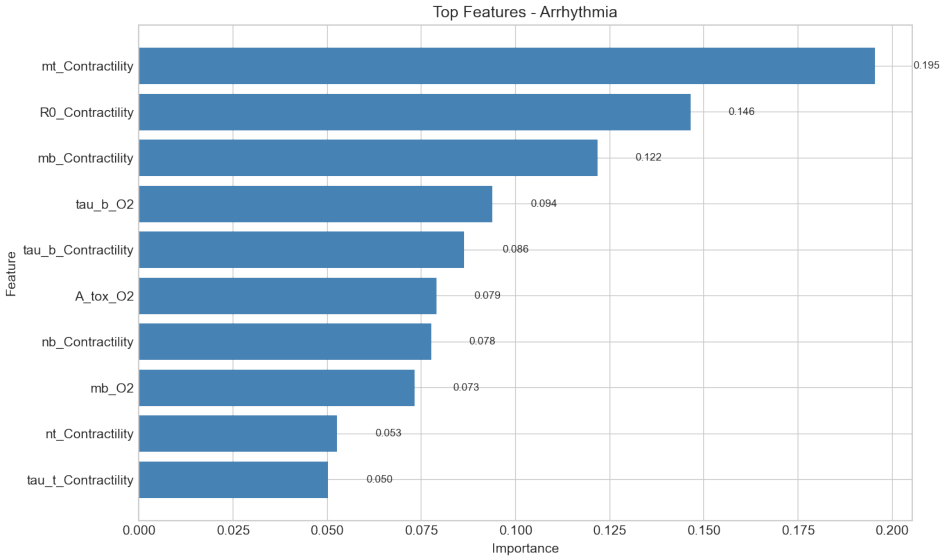

Figure 8: Top features for Arrhythmia prediction.

5.2 Heart Damage

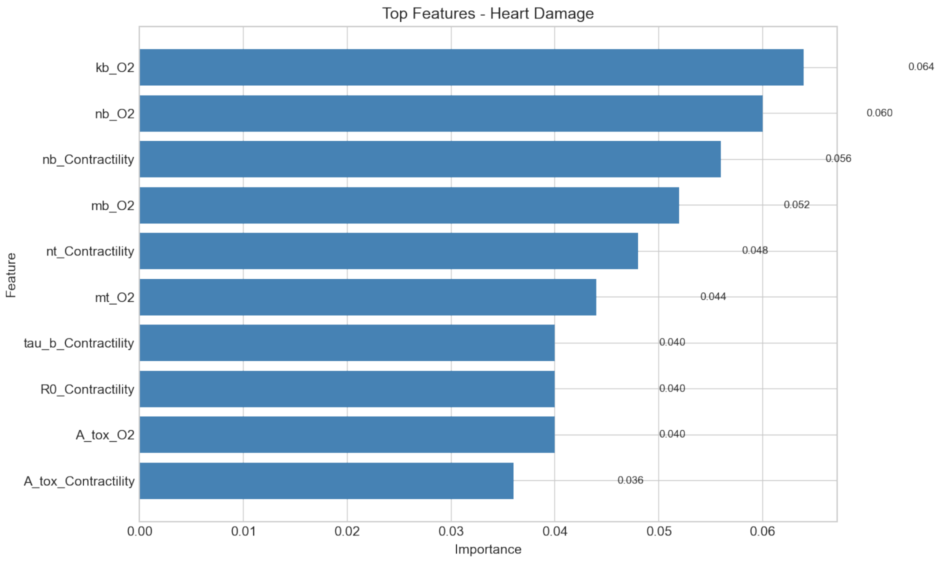

Figure 9: Top features for Heart Damage prediction.

5.3 Concern

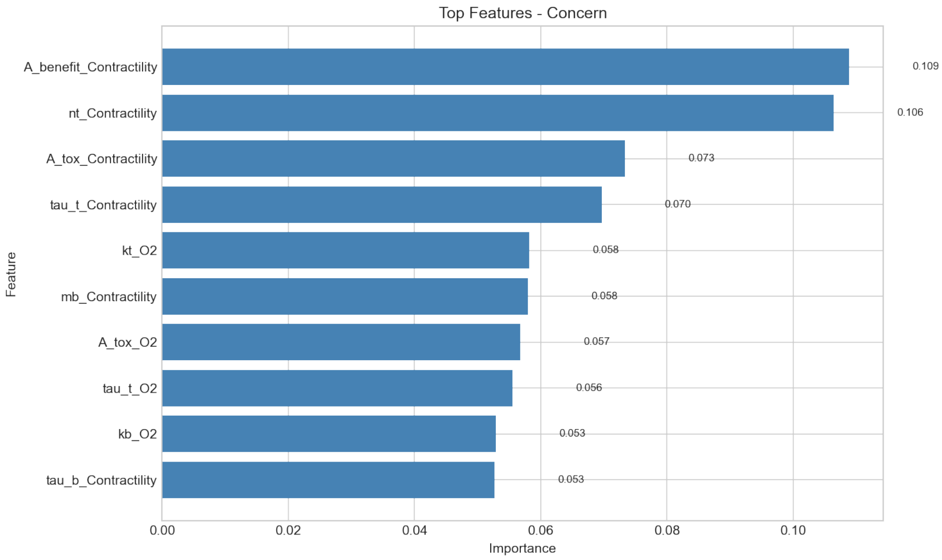

Figure 10: Top features for Concern prediction.

6 Methods

6.1 Data

Features were extracted from Dual Exponential coefficients fitted to drug response data. Each drug has coefficients for both Contractility and O2 responses.

6.2 Models

- **Arrhythmia (XGBoost)**: Gradient boosting with 100 trees, max depth 3, learning rate 0.1
- **Heart Damage (RBF SVM)**: Radial basis function kernel with C=1.0, gamma='scale'
- **Concern (Random Forest)**: 100 trees, max depth 5, min samples split 5

6.3 Cross-Validation

Leave-One-Out Cross-Validation (LOOCV) was used to evaluate model performance. Each sample was held out exactly once for testing while the remaining samples were used for training. This provides an unbiased estimate of generalization performance.

6.4 Feature Importance

- **XGBoost**: Native feature importance (gain-based)
- **RBF SVM**: Permutation importance (10 repeats)
- **Random Forest**: Native feature importance (impurity-based)

7 Conclusion

The models demonstrate varying levels of predictive performance across the three cardiac outcomes. The LOOCV approach ensures robust evaluation on the limited sample size.